

Chapter 2: Natural history of the disease, and pattern of spread of infectious disease

Objectives

- Recognize the natural history of disease
- Recognize the pattern of spread of infectious diseases

Natural history of disease

- The progress of a disease process in an individual over time, in the absence of intervention fig (2)

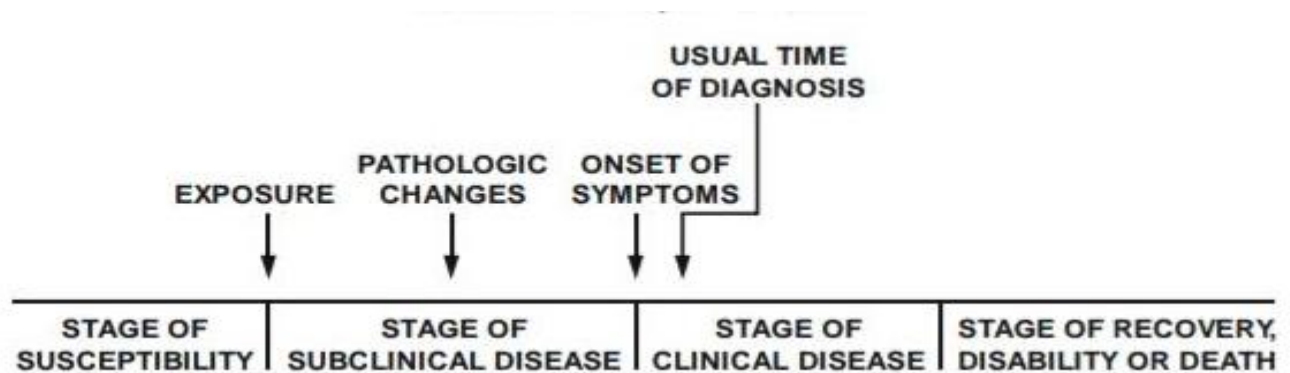


Fig. (2); Natural history of disease

- The natural history begins with the appropriate exposure to or accumulation of factors sufficient to begin the disease process in a susceptible host.
- For infectious disease, the exposure usually is a microorganism.

- For cancers, the critical factors may require both **cancer initiators**, such as asbestos fibers or components in tobacco smoke (for lung cancer) , and **cancer promoters**, such as estrogens (for endometrial cancer).

▪ **Subclinical Infection (Inapparent infection) :**

It is a state when there is a host immune response following entry of the infectious agent; the agent may also multiply in the host body, but there are no clinical manifestations of the disease.

- Although disease is inapparent during the incubation period, some pathologic changes may be detectable with laboratory, radiographic, or other screening methods.
- Most screening programs attempt to identify the disease process during this phase of its natural history, since early intervention may be more effective than treatment at a later stage of disease progression.

▪ **Incubation or latency period**

Usually, a period of subclinical or inapparent pathologic changes follows exposure, ending with the onset of symptoms.

- ✓ **For infectious diseases**, this period is usually called the **incubation period**
- ✓ **for chronic diseases**, this period is usually called **the latency period**



- **Stage of clinical disease**

- The onset of symptoms marks the transition from subclinical to clinical disease. Most diagnosis is made during the stage of clinical disease.
- In some people, however, the disease process may never progress to clinically apparent illness. In others, the disease process may result in a wide spectrum of clinical illness, ranging from mild to severe or fatal.

- **Stage of recovery, disability or death**

- The infectious diseases have various outcomes according to the infectivity, pathogenicity and virulence of the causative agent.
 - a) Infectivity refers to the proportion of exposed persons who become infected.
 - b) Pathogenicity refers to the proportion of infected persons who develop clinical disease.
 - c) Virulence refers to the proportion of persons with clinical disease who become severely ill or die.
- For example, hepatitis A virus in children has low pathogenicity and low virulence, since many infected children remain asymptomatic, and few develop severe illness.
- Rabies virus is both highly pathogenic and virulent, since virtually 100% of all infected persons (who do not receive treatment) progress to clinical disease and death.

Pattern of Spread of infectious diseases

The amount of a particular disease that is usually present in a community is the baseline level of the disease. This level is not necessarily the preferred level, which should in fact be zero; rather it is the observed level. Theoretically, if no intervention occurred and if the level is low enough not to deplete the pool of susceptible persons, the disease occurrence should continue at the baseline level indefinitely. Thus, the baseline level is often considered the expected level of the disease.

3- Endemic spread:

- a disease constantly present in the community due to maintained infection by existing ecological factors (agent, host, and environment).

Different diseases, in different communities, show different patterns of expected occurrence:

- a persistent level of occurrence with a low to moderate disease level is referred to as an endemic level. Ex., Bilharziasis is endemic in Egypt.
- a persistently high level of occurrence is called a hyperendemic level.

- an irregular pattern of occurrence, with occasional cases occurring at irregular intervals is called sporadic.

4- **Epidemic spread:**

- Increase number of cases significantly more than the usual pattern of spread of the disease. Epidemic is characterized that all the cases are interrelated :
- Time: all cases appear within a certain period of time.
- Place: all cases are reported within the same area or community.

Epidemic is either :

5- endemic infectious disease in a community and show epidemic wave at any time under adverse circumstances. E.g., Typhoid fever

B- some infectious diseases not known before or have been eliminated may appear or reappear and show epidemic wave .e.g. cholera

An epidemic may result from the following:

- a. a recent increase in amount or virulence of the agent
- b. the recent introduction of the agent into a setting where it has not been before
- c. an enhanced mode of transmission so that more susceptibles are exposed
- d. some change in the susceptibility of the host response to the agent
- e. factors that increase host exposure or involve introduction through new portals of entry

Outbreak spread: is a localized epidemic that involves a confined group or closed community as camp, school, nursery.

6- **Pandemic spread:**

- an epidemic of a particular infectious disease that spreads in between countries .e.g., COVID-19

7- **Enzootic spread:**

- endemic spread of infectious diseases in animals with the potential risk of transmission to man . e.g., tuberculosis

8- **Epizootic:**

- epidemic spread of infectious diseases in animals with the potential risk of transmission to man .e.g., foot and mouth disease.

Exercise; The case-report pattern of disease X for three communities. The communities have the same size population. Identify which term A-D below best describes the occurrence of disease X.

A. Endemic B. Epidemic C. Hyperendemic D. Pandemic

1. Community A: usually 10 cases/week; last week, 28 cases (....)

2. Community B: 50-70 cases/week; last week, 55 cases (....)

3. Community C: usually 25 cases/week; last week, 28 cases (....)